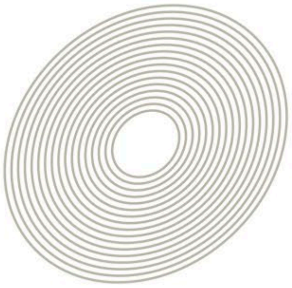
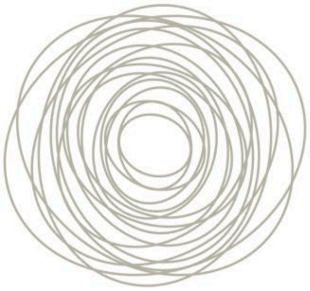
FORMATION OF SPIRAL ARMS



◁ **Perfectly ordered orbits**
In an ideal situation, objects in elliptical orbits around a galaxy's center would have their longest axes in perfect alignment with each other. Objects naturally move more slowly at the outer edges of their orbit when they are farther from the center.



◁ **Chaotic orbits**
In a completely chaotic scenario, the orbits of objects within a galaxy would be aligned in a range of different directions, and no spiral structure would form. The example here shows the same number of orbits as the other two (left and right).

◁ **Density wave**
Spiral structure arises when orbits are pulled into alignments that offset slightly from one another, often due to tidal forces from another galaxy. As a result, orbits slow down and material packs together in spiral-shaped areas of higher density.